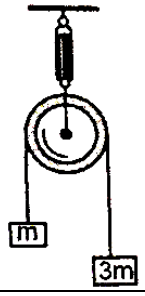
Question	Distance of the center of mass of a solid uniform cone from its vertex is Z_0. radius of its base is R and its height is h the Z_0 is equal to.
Type	multiple choice
Option	$\frac{h^2}{4R}$
Option	$\frac{3h}{4}$
Option	$\frac{5h}{8}$
Option	$\frac{3h^2}{8R}$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The center of mass of a system of two particles divides the distance between them.
Type	multiple choice
Option	In inverse ratio of square of masses of particles
Option	In direct ratio of square of masses of particles
Option	In inverse ratio of masses of particles
Option	In direct ratio of masses of particle
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Consider a system of two particles having masses m_1 and m_2. If m_1 is pushed towards the center of mass of particles through a distance d, by what distance would be particle of mass m_2 move so as to keep the center of mass of particles at the original position.
Type	multiple choice
Option	$\frac{m_1}{m_1+m_2}d$
Option	$\frac{m_1}{m_2}d$
Option	d
Option	$\frac{m_2}{m_1}d$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	If linear density of a rod of length $3m$ varies as $\lambda = 2 + x,$ then the position of the center of mass of the rod is
Type	multiple choice
Option	$\frac{7}{3}m$
Option	$\frac{12}{7}m$

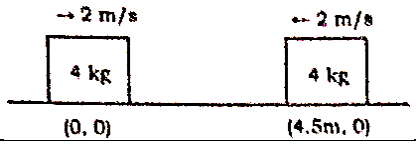
Option	$\frac{10}{7}m$
Option	$\frac{9}{7}m$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Three identical spheres, each of mass 1kg are kept as shown in figure, touching each other, with their center on a straight line. If their centers are marked P, Q, R respectively, the distance of center of mass of the system from P is
Type	multiple choice
Option	$\frac{PQ+PR+QR}{3}$
Option	$\frac{PQ+PR}{3}$
Option	$\frac{PQ+QR}{3}$
Option	$\frac{PR+QR}{3}$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Two bodies of mass 1kg and 3kg have position vectors $\hat{i}+2\hat{j}+\hat{k}$ and $-3\hat{i}-2\hat{j}+\hat{k}$, respectively. The centre of mass of the system has a position vector
Type	multiple choice
Option	$-2\hat{i}+2\hat{k}$
Option	$-2\hat{i}-\hat{j}+\hat{k}$
Option	$2\hat{i}-\hat{j}-\hat{k}$
Option	$-2\hat{i}+\hat{j}+\hat{k}$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Centre of mass of 3 particles 10kg, 20kg and 30kg is at (0, 0, 0). Where should a particle of mass 40kg be placed so that the combination of centre of mass will be at (3,3,3).
Type	multiple choice
Option	$(0,0,0)$
Option	$(7.5,7.5,7.5)$
Option	$(1,2,3)$

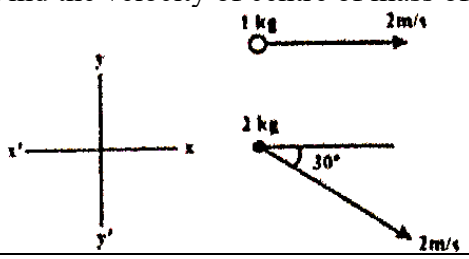
Option	$(4,4,4)$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Four point masses P,Q,R and S with respective masses 1kg, 1kg, 2kg and 2kg form the corners of a square of side a. The Centre of mass of the system will be farthest from
Type	multiple choice
Option	P only
Option	R and S
Option	R only
Option	P and Q
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A circular disc of radius R is removed from a bigger circular disc of radius 2R such that the circumferences of the discs coincide. The centre of mass of the new disc is αR from the centre of the bigger disc. The value of α is.
Type	multiple choice
Option	$1/3$
Option	$1/2$
Option	$1/6$
Option	$1/4$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Three masses are placed on the $x-axis,$ 300g at origin, 500g at $x=40\ cm$ and 400g at $x=70cm.$ The distance of the centre of mass from the origin is.
Type	multiple choice
Option	40 cm
Option	45 cm
Option	50 cm
Option	30 cm
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A stationary particle explodes into two particles of masses m_1 and m_2 which move in opposite directions with velocities v_1 and v_2. The ratio of their kinetic energies E_1/E_2 is:
Type	multiple choice
Option	m_2/m_1
Option	m_1/m_2
Option	1
Option	m_1v_2/m_2v_1
Answer	1
Solution	
Positive Marks	4

Negative Marks	1
Question	A bomb of mass 3.0 kg explodes in air into two pieces of masses 2.0 kg and 1.0 kg. The smaller mass goes at a speed of 80 m/s. The total energy imparted to the two fragments is:-
Type	multiple choice
Option	1.07 kJ
Option	2.14 kJ
Option	2.4 kJ
Option	4.8 kJ
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A bullet of mass m is fired from a gun of mass M. The recoiling gun compresses a spring of force constant k by a distance d. Then the velocity of the bullet is:- $(M >> m)$.
Type	multiple choice
Option	$kd\sqrt{M/m}$
Option	$\frac{d}{M}\sqrt{km}$
Option	$\frac{d}{m}\sqrt{kM}$
Option	$\frac{kM}{m}\sqrt{d}$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A heavy nucleus at rest breaks into two fragments which fly off with velocities 8:1. The ratio of radii of the fragments is:-
Type	multiple choice
Option	$1:2$
Option	$1:4$
Option	$4:1$
Option	$2:1$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	What will be the magnitude of recoil speed of a tank which weighs 0.4 tonnes and fires a projectile of mass 3kg with a velocity of 24 m/s?
Type	multiple choice
Option	0
Option	0.18
Option	5
Option	18
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The law of conservation of momentum is applicable for

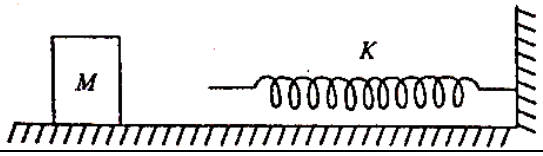
Type	multiple choice
Option	Only two particles
Option	Individual particles only
Option	A system of particles
Option	More than three particles only
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following is/are example of momentum conservation?
Type	multiple choice
Option	Motion of rockets
Option	System of gun and bullet
Option	Explosion of the bomb
Option	All of the above examples are of momentum conservation
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A bomb of mass 9kg explodes into two pieces of 3kg and 6kg. The velocity of 3kg piece is 16 m/s. The kinetic energy of 6kg piece is:-
Type	multiple choice
Option	786 J
Option	785 J
Option	192 J
Option	687 J
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A bomb initially at rest explodes by itself into three equal mass fragments. The velocities of two fragments are $\left(3\hat{i}+2\hat{j}\right)m/s$ and $\left(-\hat{i}-4\hat{j}\right)m/s$. The velocity of the third fragment is (in m/s):-
Type	multiple choice
Option	$2\hat{i}+2\hat{j}$
Option	$2\hat{i}-2\hat{j}$
Option	$-2\hat{i}+2\hat{j}$
Option	$-2\hat{i}-2\hat{j}$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A bomb of mass $m=1kg$ thrown vertically upwards with a speed $u=100m/s$ explodes into two parts after $t=5s$. A fragment of mass $m_1=400g$ moves downwards with a speed $v_1=25m/s$, then speed v_2 and direction of another mas m_2 will be:
Type	multiple choice
Option	40 m/s downwards
Option	40 m/s upwards
Option	60 m/s upwards

Option	100 m/s upwards
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	If the system is released, then the acceleration of the centre of mass of the system is :- 
Type	multiple choice
Option	$\frac{g}{4}$
Option	$\frac{g}{2}$
Option	g
Option	$2g$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Two objects of masses 200 gram and 500 gram possess velocities $10\hat{i}\ m/s$ and $3\hat{i}+5\hat{j}\ m/s$ respectively. The velocity of their center of mass in m/s is:
Type	multiple choice
Option	$5\hat{i}-25\hat{j}$
Option	$\frac{5}{7}\hat{i}-25\hat{j}$
Option	$5\hat{i}+\frac{25}{7}\hat{j}$
Option	$25\hat{i}-\frac{5}{7}\hat{j}$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	In a system of two identical particles, one particle is at rest and other has acceleration 'a'. What is the acceleration of the center of mass?
Type	multiple choice
Option	a
Option	$a/2$
Option	$2a$
Option	Zero
Answer	2
Solution	
Positive Marks	4

Negative Marks	1
Question	Two bodies having different masses 2kg and 4 kg arę moving with velocities 2 m/s and 10 m/s towards each other due to mutual gravitational attraction. Then the velocity of the centre of the mass is
Type	multiple choice
Option	$4m/s$
Option	$2m/s$
Option	$8m/s$
Option	$6m/s$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A 2kg body and a 3 kg are moving along the x-axis. At a particular instant the 2 kg body has a velocity of 3 m/s and the 3 kg body has the velocity of 2 m/s. The velocity of the centre of mass at that instant is:
Type	multiple choice
Option	$5m/s$
Option	$1m/s$
Option	0
Option	$\frac{12}{5}m/s$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Two particles of mass 2 kg and 4 kg, are projected from the top of a tower simultaneously, such that 2 kg mass is projected with a speed 20 m/s at an angle 30° above horizontal and 4 kg mass is projected at 40 m/s horizontally. The magnitude of the acceleration of the center of mass of the system of two particles will be:
Type	multiple choice
Option	$\frac{g}{2}$
Option	$\frac{g}{4}$
Option	g
Option	$2g$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	At t =0, the positions of the two blocks are shown. There is no external force acting on the system. Find the coordinates of the center of mass of the system at t = 3 seconds: 
Type	multiple choice

Option	$(1,0)$
Option	$(3,0)$
Option	$(4.5,0)$
Option	$(2.25,0)$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A boat of length 10 m and mass 450 kg is floating without motion in still water. A man of mass 50 kg standing at one end of it walks to the other end of it and stops. The magnitude of the displacement of the boat in meters relative to ground is
Type	multiple choice
Option	Zero
Option	1 m
Option	2 m
Option	5 m
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	A projectile of mass 5 kg, in its course of motion explodes on its own into two fragments. One fragment of mass 3 kg falls at $\frac{3}{4}th$ of the range.Where will the other fragment fall?
Type	multiple choice
Option	$\frac{13R}{8}$
Option	$\frac{11R}{8}$
Option	$\frac{5R}{8}$
Option	$\frac{7R}{8}$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Find the velocity of centre of mass of the system shown in the figure 
Type	multiple choice
Option	$\left(\frac{2+2\sqrt{3}}{3}\right)\hat{i}-\frac{2}{3}\hat{j}$

Option	$\left(\frac{2+2\sqrt{3}}{3}\right)\hat{i}+\frac{2}{3}\hat{j}$
Option	$\left(\frac{1+\sqrt{3}}{3}\right)\hat{i}-\frac{1}{3}\hat{j}$
Option	$\left(\frac{1+\sqrt{3}}{3}\right)\hat{i}+\frac{1}{3}\hat{j}$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A body of mass M_1 collides elastically with another mass M_2 at rest. There is maximum transfer of energy when
Type	multiple choice
Option	$M_1 > M_2$
Option	$M_1 < M_2$
Option	$M_1 = M_2$
Option	Same for all values of M_1 and M_2
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Two equal masses m_1 and m_2 moving along the same straight line with velocities $+3\,m/s$ and $-5\,m/s$ respectively collide elastically. Their velocities after the collision will be respectively
Type	multiple choice
Option	+4m/s for both
Option	-3 m/s and +5 m/s
Option	-4 m/s and +4 m/s
Option	-5 m/s and +3 m/s
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A body of mass 2 kg makes an elastic with another body at rest and continues to move in the original direction with one fourth of its original speed. The mass of the second body which collides with the first body is
Type	multiple choice
Option	2 kg
Option	1.2 kg
Option	3 kg
Option	1.5 kg
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	In an inelastic collision
Type	multiple choice
Option	energy is conserved
Option	energy is not conserved

Option	momentum is not conserved
Option	none of the above
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	A ball dropped from a height of 2m rebounds to a height of 1.5m after hitting the ground. Then the percentage of energy lost is
Type	multiple choice
Option	25
Option	30
Option	50
Option	100
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	The block of mass M moving on the frictionless horizontal surface collides with the spring of spring constant k and compress it by length L. The maximum momentum of the block after collision is 
Type	multiple choice
Option	$\frac{kL^2}{2M}$
Option	$L\sqrt{Mk}$
Option	$\frac{ML^2}{k}$
Option	Zero
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	A ball is moving with a speed of 5 m/s and collides with another ball moving with a speed 1 m/s in the opposite direction. What will be the speed of the second ball if after the collision the first ball comes to rest and the coefficient of 12 restitution is $\frac{1}{3}$?
Type	multiple choice
Option	2 m/s
Option	3 m/s
Option	4 m/s
Option	$\frac{4}{3} m/s$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A lead ball strikes a wall and falls down, a tennis ball having the same mass and

	velocity strikes the wall and bounces back. Check the correct statement.
Type	multiple choice
Option	The lead ball suffers a greater change in momentum compared with the tennis ball
Option	The tennis ball suffers a greater change in momentum as compared with the lead ball
Option	Both suffer an equal change in momentum
Option	None of the above
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	A body of mass m moving with velocity v collides head on elastically with another body of mass 2m which is initially at rest. The ratio of K.E. of colliding body of mass m before and after collision will be
Type	multiple choice
Option	$1:1$
Option	$2:1$
Option	$4:1$
Option	$9:1$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A tennis ball is released from height h above ground level. If the ball makes inelastic collision with the ground, to what height will it rise after first collision (e is coefficient of restitution)
Type	multiple choice
Option	he^6
Option	e^2h
Option	e^3h
Option	None of these
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	A bullet of mass m moving with velocity v strikes a block of mass M at rest and gets embedded into it. The kinetic energy of the composite block will be
Type	multiple choice
Option	$\frac{1}{2}mv^2 \times \frac{M}{(m+M)}$
Option	$\frac{1}{2}mv^2 \times \frac{m}{(m+M)}$
Option	$\frac{1}{2}mv^2 \times \frac{(M+m)}{M}$
Option	$\frac{1}{2}Mv^2 \times \frac{m}{(m+M)}$
Answer	2
Solution	
Positive Marks	4

Negative Marks	1
Question	A particle of mass m_1 moving with u_1 velocity collide with another particle of mass m_2 which is initially in rest. If collision is perfectly inelastic then the fractional loss in the kinetic energy is
Type	multiple choice
Option	$\frac{1}{4}m_1u_1^2$
Option	$\frac{m_2}{m_1}$
Option	$\frac{m_2}{m_1+m_2}$
Option	$\frac{m_1}{m_1+m_2}$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A ball moving with velocity 2 m/s collides head on with another stationary ball of double the mass. If the coefficient of restitution is 0.5. then their velocities (in m/s) after collisions will be
Type	multiple choice
Option	$1, 0.5$
Option	$0, 1$
Option	$1, 1$
Option	$0, 2$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	A ball of mass m moving with a constant velocity strike against a ball of same mass at rest. If e = coefficient of restitution., then what will be the ratio of velocity of two balls after collision?
Type	multiple choice
Option	$\frac{1-e}{1+e}$
Option	$\frac{e-1}{e+1}$
Option	$\frac{1+e}{1-e}$
Option	$\frac{2+e}{e-1}$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A mass of 10 gm moving with a velocity of 100 cm/s strikes a pendulum bob of mass 10 gm. The two masses stick together. The maximum height reached by the system now is $\left(g = 10\,m/s^2\right)$

Type	multiple choice
Option	Zero
Option	5 cm
Option	2.5 cm
Option	1.25 cm
Answer	4
Solution	
Positive Marks	4
Negative Marks	1